Exercise 13.2 SOLUTION

Q.1. Differentiate w.r.t' x' .

i. $x^4+2x^3+x^2$

Sol: Let $y = x^4 + 2x^3 + x^2$

$$\frac{dy}{dx}=\frac{d}{dx}x^4+2\frac{d}{dx}x^3+\frac{d}{dx}x^2=4x^3+2.3x^2+2x$$

$$\frac{dy}{dx} = 4x^3 + 6x^2 + 2x.$$

ii. $x^{-3}+2x^{-\frac{3}{2}}+3$

Sol: Let $y = x^{-3} + 2x^{-3/2}+3$

$$\frac{dy}{dx}=\frac{d}{dx}x^{-3}+2\frac{d}{dx}x^{-3/2}+\frac{d}{dx}3=-3x^{-4}+2\left(-\frac{3}{2}\right)x^{-3/2-1}+0$$

$$\frac{dy}{dx}=-3x^{-4}-3x^{-5/2}=\frac{-3}{x^4}-\frac{3}{x^{5/2}}=-3\left(\frac{1}{x^4}+\frac{1}{x^{5/2}}\right)$$

iii. $\dfrac{2x-3}{2x+1}$

Sol: Let $y=\dfrac{2x-3}{2x+1}\Rightarrow\dfrac{dy}{dx}=\dfrac{d}{dx}\left(\dfrac{2x-3}{2x+1}\right)$

$$\frac{dy}{dx}=\frac{(2x+1)\frac{d}{dx}(2x-3)-(2x-3)\frac{d}{dx}(2x+1)}{(2x+1)^2}=\frac{(2x+1)(2-0)-(2x-3)(2+0)}{(2x+1)^2}$$

$$\frac{dy}{dx}=\frac{\cancel{4x}+2-\cancel{4x}+6}{(2x+1)^2}=\frac{8}{(2x+1)^2}$$

iv. $\dfrac{\left(1+\sqrt{x}\right)\left(x-x^{\frac{3}{2}}\right)}{\sqrt{x}}$

Sol: Let $y=\dfrac{\left(1+\sqrt{x}\right)\left(x-x^{3/2}\right)}{\sqrt{x}}$

$$y=\frac{x\left(1+\sqrt{x}\right)\left(1-x^{1/2}\right)}{\sqrt{x}}=\frac{\cancel{\sqrt{x}}\times\sqrt{x}\left(1+\sqrt{x}\right)\left(1-\sqrt{x}\right)}{\cancel{\sqrt{x}}}$$

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$$y = \sqrt{x}\left((1)^2 - \left(\sqrt{x}\right)^2\right) = \sqrt{x}(1-x) = (x)^{1/2} - x^{1/2+1}$$

$$y = x^{1/2} - x^{3/2} \Rightarrow \frac{dy}{dx} = \frac{d}{dx}x^{1/2} - \frac{d}{dx}x^{3/2}$$

$$\frac{dy}{dx} = \frac{1}{2}x^{1/2-1} - \frac{3}{2}x^{3/2-1} = \frac{1}{2}x^{-1/2} - \frac{3}{2}x^{1/2} = \frac{1}{2\sqrt{x}} - \frac{3\sqrt{x}}{2}$$

$$\frac{dy}{dx} = \frac{1-3x}{2\sqrt{x}}$$

v. $\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2$

Sol: Let $y = \left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 = \left(\sqrt{x}\right)^2 + \left(\frac{1}{\sqrt{x}}\right)^2 - 2\sqrt{x}\frac{1}{\sqrt{x}}$

$$y = x + \frac{1}{x} - 2 \quad \Rightarrow \quad y = x + x^{-1} - 2 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{d}{dx}(x) + \frac{d}{dx}x^{-1} - \frac{d}{dx}(2)$$

$$\frac{dy}{dx} = 1 + (-1)x^{-2} - 0 = 1 - \frac{1}{x^2} = \frac{x^2-1}{x^2}$$

vi. $(x-5)(3-x)$

Sol: Let $y = (x-5)(3-x) \Rightarrow \frac{dy}{dx} = \frac{d}{dx}(x-5)(3-x)$

$$\frac{dy}{dx} = (x-5)\frac{d}{dx}(3-x) + (3-x)\frac{d}{dx}(x-5) = (x-5)(0-1) + (3-x)(1-0)$$

$$\frac{dy}{dx} = -x + 5 + 3 - x = 8 - 2x$$

vii. $\frac{\left(x^2+1\right)^2}{x^2-1}$

Sol: Let $y = \frac{\left(x^2+1\right)^2}{x^2-1}$

$$\frac{dy}{dx} = \frac{\left(x^2-1\right)\frac{d}{dx}\left(x^2+1\right)^2 - \left(x^2+1\right)^2\frac{d}{dx}\left(x^2-1\right)}{\left(x^2-1\right)^2}$$

$$\frac{dy}{dx} = \frac{\left(x^2-1\right)2\left(x^2+1\right)^1(2x) - \left(x^2+1\right)^2 2x}{\left(x^2-1\right)^2}$$

$$\frac{dy}{dx} = \frac{2x\left(x^2+1\right)\left[2\left(x^2-1\right) - \left(x^2+1\right)\right]}{\left(x^2-1\right)^2}$$

$$\frac{dy}{dx} = \frac{2x\left(x^2+1\right)\left(2x^2-2-x^2-1\right)}{\left(x^2-1\right)^2} = \frac{2x(x^2+1)(x^2-3)}{(x^2-1)^2}$$

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viii. $\frac{x^2+1}{x^2-3}$

Sol: Let $y = \frac{x^2+1}{x^2-3}$.

$$\frac{dy}{dx} = \frac{(x^2-3)\frac{d}{dx}(x^2+1)-(x^2+1)\frac{d}{dx}(x^2-3)}{(x^2-3)^2} = \frac{(x^2-3)(2x)-(x^2+1)(2x)}{(x^2-3)^2} = \frac{2x(x^2-3-x^2-1)}{(x^2-3)^2}$$

$$\frac{dy}{dx} = \frac{2x(-4)}{(x^2-3)^2} = \frac{-8x}{(x^2-3)^2}$$

ix. $\frac{2x-1}{\sqrt{x^2+1}}$

Sol: Let $y = \frac{2x-1}{\sqrt{x^2+1}}$

$$\frac{dy}{dx} = \frac{(x^2+1)^{1/2}\frac{d}{dx}(2x-1)-(2x-1)\frac{d}{dx}(x^2+1)^{1/2}}{\left[(x^2+1)^{1/2}\right]^2}$$

$$\frac{dy}{dx} = \frac{\sqrt{(x^2+1)}(2)-(2x-1)\frac{1}{2}(x^2+1)^{-1/2}(2x)}{(x^2+1)}$$

$$\frac{dy}{dx} = \left[2\sqrt{(x^2+1)}-\frac{x(2x-1)}{\sqrt{x^2+1}}\right]\frac{1}{(x^2+1)}\frac{1}{(x^2+1)} = \left[\frac{2(x^2+1)-(2x^2-x)}{\sqrt{x^2+1}}\right] = \frac{2x^2+2-2x^2+x}{(x^2+1)(x^2+1)^{1/2}}$$

$$\frac{dy}{dx} = \frac{x+2}{(x^2+1)^{3/2}}$$

x. $\sqrt{\frac{a-x}{a+x}}$

Sol: Let $y = \sqrt{\frac{a-x}{a+x}} = \frac{(a-x)^{1/2}}{(a+x)^{1/2}}$

$$\frac{dy}{dx} = \frac{(a+x)^{1/2}\frac{d}{dx}(a-x)^{1/2}-(a-x)^{1/2}\frac{d}{dx}(a+x)^{1/2}}{\left[(a+x)^{1/2}\right]^2} = \frac{(a+x)^{1/2}\frac{1}{2}(a-x)^{-1/2}(-1)-(a-x)^{1/2}\frac{1}{2}(a+x)^{-1/2}(1)}{(a+x)}$$

$$\frac{dy}{dx} = \left[\sqrt{a+x}\frac{-1}{2\sqrt{a-x}}-\frac{\sqrt{a-x}}{2\sqrt{a+x}}\right]\frac{1}{(a+x)} = \frac{1}{(a+x)}\left[\frac{-(a+x)-(a-x)}{2\sqrt{a-x}\sqrt{a+x}}\right] = \frac{-a-x-a+x}{2\sqrt{(a-x)}(a+x)^{3/2}}$$

$$\frac{dy}{dx} = \frac{-2a}{2\sqrt{a-x}(a+x)^{3/2}} = \frac{-a}{\sqrt{(a-x)}(a+x)^{3/2}}$$

xi. $\frac{\sqrt{x^2+1}}{\sqrt{x^2-1}}$

Sol: Let $y = \frac{\sqrt{x^2+1}}{\sqrt{x^2-1}} = \frac{(x^2+1)^{1/2}}{(x^2-1)^{1/2}}$

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$$\frac{dy}{dx}=\frac{\left(x^2-1\right)^{1/2}\frac{d}{dx}\left(x^2+1\right)^{1/2}-\left(x^2+1\right)^{1/2}\frac{d}{dx}\left(x^2-1\right)^{1/2}}{\left[\left(x^2-1\right)^{1/2}\right]^2}$$

$$\frac{dy}{dx}=\frac{1}{\left(x^2-1\right)}\left[\sqrt{x^2-1}\,\frac{1}{2}\left(x^2+1\right)^{-1/2}(2x)-\sqrt{x^2+1}\,\frac{1}{2}\left(x^2-1\right)^{-1/2}(2x)\right]$$

$$\frac{dy}{dx}=\frac{1}{\left(x^2-1\right)}\left[\frac{x\sqrt{x^2-1}}{\sqrt{x^2+1}}-\frac{x\sqrt{x^2+1}}{\sqrt{x^2-1}}\right]=\frac{1}{\left(x^2-1\right)}\left[\frac{x\left(x^2-1\right)-x\left(x^2+1\right)}{\sqrt{x^2+1}\sqrt{x^2-1}}\right]=\frac{x}{\left(x^2-1\right)}\left[\frac{x^2-1-x^2-1}{\sqrt{x^2+1}\left(x^2-1\right)^{1/2}}\right]=\frac{-2x}{\sqrt{x^2+1}\left(x^2-1\right)^{3/2}}$$

Q.2. Find $\frac{dy}{dx}$ if $y=\dfrac{\left(\sqrt{x}+1\right)\left(x^{\frac{3}{2}}-1\right)}{x^{\frac{1}{2}}-1}$, $(x\neq 1)$

Sol: Given that $y=\dfrac{\left(\sqrt{x}+1\right)\left(x^{\frac{3}{2}}-1\right)}{x^{\frac{1}{2}}-1},(x\neq 1)$

$$y=\frac{x^2-\sqrt{x}+x^{\frac{3}{2}}-1}{\sqrt{x}-1} \qquad \left(\sqrt{x}\times x^{\frac{3}{2}}=x^{\frac{1}{2}+\frac{3}{2}}=x^{\frac{4}{2}}=x^2\right)$$

Differentiating w.r.t "x"

$$\frac{d}{dx}(y)=\frac{d}{dx}\left(\frac{x^2-\sqrt{x}+x^{\frac{3}{2}}-1}{\sqrt{x}-1}\right)$$

$$\frac{dy}{dx}=\frac{\left(\sqrt{x}-1\right)\frac{d}{dx}\left(x^2-\sqrt{x}+x^{\frac{3}{2}}-1\right)-\left(x^2-\sqrt{x}+x^{\frac{3}{2}}-1\right)\frac{d}{dx}\left(\sqrt{x}-1\right)}{\left(\sqrt{x}-1\right)^2}$$

$$=\frac{\left(\sqrt{x}-1\right)\left(2x-\frac{1}{2\sqrt{x}}+\frac{3}{2}x^{\frac{1}{2}}\right)-\left(x^2-\sqrt{x}+x^{\frac{3}{2}}-1\right)\left(\frac{1}{2\sqrt{x}}\right)}{\left(\sqrt{x}-1\right)^2}$$

$$=\frac{\left(\sqrt{x}-1\right)\left(\frac{4x\sqrt{x}-1+3x}{2\sqrt{x}}\right)-\frac{1}{2\sqrt{x}}\left(x^2-\sqrt{x}+x^{\frac{3}{2}}-1\right)}{\left(\sqrt{x}-1\right)^2}$$

$$=\frac{\frac{1}{2\sqrt{x}}\left\{\left(\sqrt{x}-1\right)\left(4x^{\frac{3}{2}}-1+3x\right)-\left(x^2-\sqrt{x}+x^{\frac{3}{2}}-1\right)\right\}}{\left(\sqrt{x}-1\right)^2}=\frac{4x^2-\sqrt{x}+3x^{\frac{3}{2}}-4x^{\frac{3}{2}}+1-3x-x^2+\sqrt{x}-x^{\frac{3}{2}}+1}{2\sqrt{x}\left(\sqrt{x}-1\right)^2}$$

$$\frac{dy}{dx}=\frac{3x^2-2x^{\frac{3}{2}}-3x+2}{2\sqrt{x}\left(\sqrt{x}-1\right)^2}$$

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Q.3. Differentiate $\dfrac{\left(\sqrt{x}+1\right)\left(x^{\frac{3}{2}}-1\right)}{x^{\frac{3}{2}}-x^{\frac{1}{2}}}$ with respect to x.

Sol: Given that $y=\dfrac{\left(\sqrt{x}+1\right)\left(x^{\frac{3}{2}}-1\right)}{x^{\frac{3}{2}}-x^{\frac{1}{2}}}$

$$y=\frac{x^2-\sqrt{x}+x^{\frac{3}{2}}-1}{x^{\frac{3}{2}}-x^{\frac{1}{2}}}$$

Differentiating w.r.t. 'x'

$$\frac{d}{dx}(y)=\frac{d}{dx}\left(\frac{x^2-\sqrt{x}+x^{\frac{3}{2}}-1}{x^{\frac{3}{2}}-x^{\frac{1}{2}}}\right)$$

$$\frac{dy}{dx}=\frac{\left(x^{\frac{3}{2}}-x^{\frac{1}{2}}\right)\frac{d}{dx}\left(x^2-\sqrt{x}+x^{\frac{3}{2}}-1\right)-\left(x^2-\sqrt{x}+x^{\frac{3}{2}}-1\right)\frac{d}{dx}\left(x^{\frac{3}{2}}-x^{\frac{1}{2}}\right)}{\left(x^{\frac{3}{2}}-x^{\frac{1}{2}}\right)^2}$$

$$=\frac{\left(x^{\frac{3}{2}}-x^{\frac{1}{2}}\right)\left(2x-\frac{1}{2\sqrt{x}}+\frac{3}{2}x^{\frac{1}{2}}\right)-\left(x^2-\sqrt{x}+x^{\frac{3}{2}}-1\right)\left(\frac{3}{2}x^{\frac{1}{2}}-\frac{1}{2}x^{-\frac{1}{2}}\right)}{\left(x^{\frac{3}{2}}-x^{\frac{1}{2}}\right)^2}$$

$$=\frac{\left(x^{\frac{3}{2}}-x^{\frac{1}{2}}\right)\left(\frac{4x^{\frac{3}{2}}-1+3x}{2\sqrt{x}}\right)-\left(x^2-\sqrt{x}+x^{\frac{3}{2}}-1\right)\left(\frac{3x-1}{2\sqrt{x}}\right)}{\left(x^{\frac{3}{2}}-x^{\frac{1}{2}}\right)^2}$$

$$=\frac{\frac{1}{2\sqrt{x}}\left\{\left(x^{\frac{3}{2}}-x^{\frac{1}{2}}\right)\left(4x^{\frac{3}{2}}-1+3x\right)-\left(x^2-\sqrt{x}+x^{\frac{3}{2}}-1\right)\left(\frac{3x-1}{2\sqrt{x}}\right)\right\}}{\left(x^{\frac{3}{2}}-x^{\frac{1}{2}}\right)^2}$$

$$=\frac{4x^3-x^{\frac{7}{2}}+3x^{\frac{5}{2}}-4x^2+\sqrt{x}-3x^{\frac{3}{2}}-3x^3+x^2+3x^{\frac{5}{2}}-\sqrt{x}-3x^{\frac{3}{2}}+x^{\frac{7}{2}}+3x-1}{2\sqrt{x}.\left(x^{\frac{3}{2}}-x^{\frac{1}{2}}\right)^2}$$

$$\frac{dy}{dx}=\frac{x^3-3x^2+3x-1}{2\sqrt{x}.\left(x^{\frac{3}{2}}-x^{\frac{1}{2}}\right)^2}$$

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Q.4. If $y = \sqrt{x} - \frac{1}{\sqrt{x}}$, show that $2x\frac{dy}{dx} + y = 2\sqrt{x}$

Sol. $y = \sqrt{x} - \frac{1}{\sqrt{x}} = x^{1/2} - x^{-1/2}$

$$\frac{dy}{dx} = \frac{1}{2}x^{-1/2} - \left(-\frac{1}{2}\right)x^{-3/2} = \frac{1}{2\sqrt{x}} + \frac{1}{2x^{3/2}}$$

L.H.S $= 2x\frac{dy}{dx} + y = 2x\left(\frac{1}{2\sqrt{x}} + \frac{1}{2x^{3/2}}\right) + \sqrt{x} - \frac{1}{\sqrt{x}}$ (*Put value of* y *and* $\frac{dy}{dx}$)

$= \sqrt{x} + \frac{1}{\sqrt{x}} + \sqrt{x} - \frac{1}{\sqrt{x}} = 2\sqrt{x}$ Hence Proved.

Q.5. If $y = x^4 + 2x^2 + 2$, prove that $\frac{dy}{dx} = 4x\sqrt{y-1}$

Sol. Let $y = x^4+2x^2+2$ ____I

$$\frac{dy}{dx} = 4x^3+2.2x+0 = 4x^3+4x$$

$$\frac{dy}{dx} = 4x(x^2+1)\text{____II}$$

From I $y = x^4+2x^2+2$

or $y-1 = x^4+2x^2+2-1$

or $y-1 = x^4+2x^2+1$

or $y-1 = (x^2+1)^2$

or $\sqrt{y-1} = (x^2+1)$

Put value of x^2+1 in II then

$$\frac{dy}{dx} = 4x\sqrt{y-1}$$

hence proved

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